🌍 Next-Generation AI Framework for Intelligent Travel Support in Sri Lanka

An AI-powered smart tourism management platform designed to improve the travel experience in Sri Lanka through intelligent planning, crowd prediction, safety verification, and adaptive tourism technologies.

This research project focuses on solving real-world tourism challenges such as overcrowding, poor travel planning, transportation safety, and lack of real-time tourist support by integrating Artificial Intelligence, GIS technologies, Machine Learning, and real-time analytics.

🚀 Project Overview

Sri Lanka’ s tourism industry faces multiple challenges including:

- Overcrowded tourist destinations
- Lack of real-time crowd monitoring
- Transportation safety concerns
- Poor travel route optimization
- Static travel planning systems
- Limited personalized tourism support

This project introduces an intelligent AI-driven ecosystem that provides:

- Smart travel planning
- Real-time adaptive route optimization
- Crowd prediction and redistribution
- Emotion-aware tourist experience management
- Safety and transport verification
- AI-powered conversational support

🧩 System Components

1️⃣ AI Conversational Travel Analyzer + Intelligent Package Generator

👨‍💻 Member

IT22082442 – Senevirathna T.M.N.D.L

📌 Description

An AI-powered travel planning system that analyzes user preferences using NLP and Large Language Models to generate personalized travel packages and adaptive itineraries for Sri Lanka.

✨ Features

- Intelligent itinerary generation
- Real-time travel adaptation
- Voice-based travel assistant
- Weather and transport-aware rerouting
- Personalized recommendations

🛠️ Technologies

- React.js
- Node.js
- FastAPI
- MongoDB
- Llama 3 / GPT Models
- Genetic Algorithms
- REST APIs

2️⃣ Crowd Prediction & Tourism Flow Optimization System

👨‍💻 Member

IT22607614 – Anuja Manthrirathna

📌 Description

An AI + GIS-based tourism flow optimization system that predicts tourist crowd levels using real-time and historical data to reduce overcrowding and improve destination distribution across Sri Lanka.

✨ Features

- Real-time crowd heatmaps
- Crowd prediction dashboard
- Smart route optimization
- Alternative destination suggestions
- Seasonal demand analysis
- Event impact analysis

🛠️ Technologies

- React.js
- FastAPI
- Node.js
- MongoDB (GeoJSON)
- ArcGIS
- QGIS
- TensorFlow
- Scikit-learn
- Mapbox API
- OpenWeather API

🤖 ML Models & Algorithms

- LSTM
- ARIMA
- Random Forest
- K-Means Clustering
- Kernel Density Estimation (KDE)
- Dijkstra Algorithm
- A* Routing Algorithm

3️⃣ Adaptive Emotion Recognition for Tourist Experience Management

👨‍💻 Member

IT22216014 Kaweeshwara M.D.O

📌 Description

A multi-modal emotion recognition system that monitors tourist emotions and environmental context to improve visitor satisfaction and detect unusual situations in real time.

✨ Features

- Tourist emotion monitoring
- Real-time engagement heatmaps
- Satisfaction analysis
- Emergency behavior detection
- Smart recommendations for experience improvement

🛠️ Technologies

- Python
- OpenCV
- TensorFlow
- CNN Models
- LSTM Models
- React.js

4️⃣ Smart Safety & Transport Verification System

👨‍💻4 Member

IT22176974 Rupasinghe M.L.K.S

📌 Description

A smart transport safety platform that verifies tourists and drivers using Face-ID technology while monitoring travel safety using AI-powered risk analysis.

✨ Features

- Dual Face-ID verification
- GPS tracking
- AI safety risk scoring
- Emergency alert notifications
- Route adherence monitoring

🛠️ Technologies

- React.js
- Node.js
- MongoDB
- OpenCV
- GPS APIs
- AI Risk Analysis Models

🧠 Research Novelty

- Combines AI + GIS technologies
- Uses real-time and historical datasets
- Predicts future tourism crowd behavior
- Provides proactive travel assistance
- Sri Lanka-focused intelligent tourism ecosystem
- Integrates safety, personalization, and optimization into one platform

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📊 Datasets & APIs

- ArcGIS Open Data
- OpenStreetMap
- SLTDA Tourism Data

- Google Popular Times
- OpenWeather API
- Transport Datasets

💻 Technology Stack

Category	Technologies
Frontend	React.js, React Native
Backend	Node.js, Express.js, FastAPI
Database	MongoDB
GIS	ArcGIS, QGIS, Mapbox
AI / ML	TensorFlow, Scikit-learn
APIs	OpenWeather API, Google Maps API
Programming Languages	JavaScript, TypeScript, Python

📌 Expected Outputs

- AI-powered travel planning platform
- Real-time crowd prediction system
- Tourism heatmaps
- Smart recommendation engine
- Safety verification dashboard
- Adaptive tourism management system

📚 Research Domain

- Artificial Intelligence
- Smart Tourism

- Machine Learning
- GIS & Spatial Computing
- Computer Vision
- Natural Language Processing

👨‍🏫 Supervisors

Main Supervisor

- Dr. Kaushalya Dissanayake

Co-Supervisor

- Mrs. Suriya

👥 Research Team

Member Name	Registration Number	Component
Senevirathna T.M.N.D.L	IT22082442	AI Conversational Travel Analyzer
Anuja Manthrirathna	IT22607614	Crowd Prediction & Tourism Flow Optimization
Kaweeshwara M.D.O	IT22216014	Adaptive Emotion Recognition
Rupasinghe M.L.K.S	IT22176974	Smart Safety & Transport Verification

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🏫 Academic Information

University: Sri Lanka Institute of Information Technology (SLIIT)
Degree Program: BSc (Hons) in Information Technology

Research Group: SST – Software Systems & Technologies
Project ID: R26-IT-112

📂 Repository Structure

```
project-root/
│
├── frontend/
├── backend/
├── ml-models/
├── datasets/
├── assets/
├── README.md
└── package.json
```